

8255 Programmable Peripheral Interface (PPI) MPCA Module 4

> **Definition:** The **Intel 8255A (PPI)** is a general-purpose programmable I/O device designed to interface parallel peripheral devices (keyboards, displays, ADCs, printers) to the 8086 microprocessor.

1. Detailed Technical Explanation

1. Internal Block Diagram of 8255

The 8255 features **24** programmable I/O pins divided into three 8-bit ports:

8255 I

2. 8255 Port Selection Truth Table

| **CS** | **A₁** | **A₀** | Selected Register / Port |

3. Control Word Register (CWR) Formats

1. I/O Mode Set Control Word (**D₇ = 1**)

2. Operating Modes of 8255:

1. **Mode 0 (Basic I/O):** Simple input or output without handshake signals. All ports (A, B, C) operate independently.

2. **Mode 1 (Strobed I/O):** (STB, STB, STB)

4. Must-Write Points for Exams

- 8255 has 24 I/O pins configured into Port A, Port B, and Port C (Upper & Lower).

- **Mode 2 (Bidirectional I/O):**

5. Quick Recall Flow

8255 P
Mode